Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Advanced Java (01CT1623)	**Aim:** Extend Task 1 to create a web-based Library Management System using Servlets	
Take Home Exam 2	**Date :-22-04-2026**	**Enrollment No:-** 92301733024

servlet/AddBookServlet.java

```
package servlet;

import dao.BookDAO;
import model.Book;

import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.*;
import java.io.IOException;

@WebServlet("/addBook")
public class AddBookServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if (session == null || session.getAttribute("user") == null) {
            response.sendRedirect("login.jsp");
            return;
        }

        String title = request.getParameter("title");
        String author = request.getParameter("author");
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        BookDAO dao = new BookDAO();
        dao.addBook(new Book(title, author, quantity));

        response.sendRedirect("viewBooks");
    }
}
```

Loginservlet:

```
package servlet;

import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.*;
import java.io.IOException;
```

```
@WebServlet("/login")
public class LoginServlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        if ("admin".equals(username) && "admin123".equals(password)) {

            HttpSession session = request.getSession();
            session.setAttribute("user", username);

            response.sendRedirect("addBook.jsp");

        } else {
            request.setAttribute("error", "Invalid Credentials");
            RequestDispatcher rd = request.getRequestDispatcher("login.jsp");
            rd.forward(request, response);
        }
    }
}
```

ViewBookServlet.java:

```
package servlet;

import dao.BookDAO;
import model.Book;

import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.*;
import java.io.IOException;
import java.util.List;

@WebServlet("/viewBooks")
public class ViewBooksServlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        HttpSession session = request.getSession(false);

        if (session == null || session.getAttribute("user") == null) {
```

```
        response.sendRedirect("login.jsp");
        return;
    }

    BookDAO dao = new BookDAO();
    List<Book> books = dao.getAllBooks();

    request.setAttribute("bookList", books);

    RequestDispatcher rd = request.getRequestDispatcher("viewBooks.jsp");
    rd.forward(request, response);
  }
}
```

addBook.jsp:

```
<!DOCTYPE html>
<html>
<head>
  <title>Add Book</title>
  <style>
    body {
        font-family: Arial;
        background: #f4f6f9;
        margin: 0;
    }
    .navbar {
        background: #667eea;
        color: white;
        padding: 15px;
        text-align: center;
        font-size: 20px;
    }
    .container {
        width: 400px;
        margin: 40px auto;
        background: white;
        padding: 25px;
        border-radius: 10px;
        box-shadow: 0 5px 15px rgba(0,0,0,0.2);
    }
    input {
        width: 100%;
        padding: 10px;
        margin: 10px 0;
```

```
        }
        button {
            width: 100%;
            padding: 10px;
            background: #667eea;
            color: white;
            border: none;
        }
        a {
            display: block;
            margin-top: 15px;
            text-align: center;
        }
    </style>
</head>
<body>

<div class="navbar">Library Management System</div>

<div class="container">
    <h3>Add New Book</h3>

    <form action="addBook" method="post">
        <input type="text" name="title" placeholder="Book Title" required>
        <input type="text" name="author" placeholder="Author" required>
        <input type="number" name="quantity" placeholder="Quantity" required>
        <button type="submit">Add Book</button>
    </form>

    <a href="viewBooks">View Books</a>
</div>

</body>
</html>
```

Login.jsp:

```
<!DOCTYPE html>
<html>
<head>
    <title>Login</title>
    <style>
        body {
            font-family: Arial;
            background: linear-gradient(135deg, #667eea, #764ba2);
```

```
        display: flex;
        justify-content: center;
        align-items: center;
        height: 100vh;
    }
    .card {
        background: white;
        padding: 30px;
        border-radius: 10px;
        box-shadow: 0 10px 25px rgba(0,0,0,0.2);
        width: 300px;
        text-align: center;
    }
    input {
        width: 100%;
        padding: 10px;
        margin: 10px 0;
        border-radius: 5px;
        border: 1px solid #ccc;
    }
    button {
        width: 100%;
        padding: 10px;
        background: #667eea;
        color: white;
        border: none;
        border-radius: 5px;
        cursor: pointer;
    }
    button:hover {
        background: #5a67d8;
    }
    .error {
        color: red;
    }
  </style>
</head>
<body>

<div class="card">
  <h2>Library Login</h2>

  <form action="login" method="post">
    <input type="text" name="username" placeholder="Username" required>
    <input type="password" name="password" placeholder="Password" required>
    <button type="submit">Login</button>
```

```
    </form>

    <p class="error">${error}</p>
</div>

</body>
</html>
```

viewBooks.jsp:

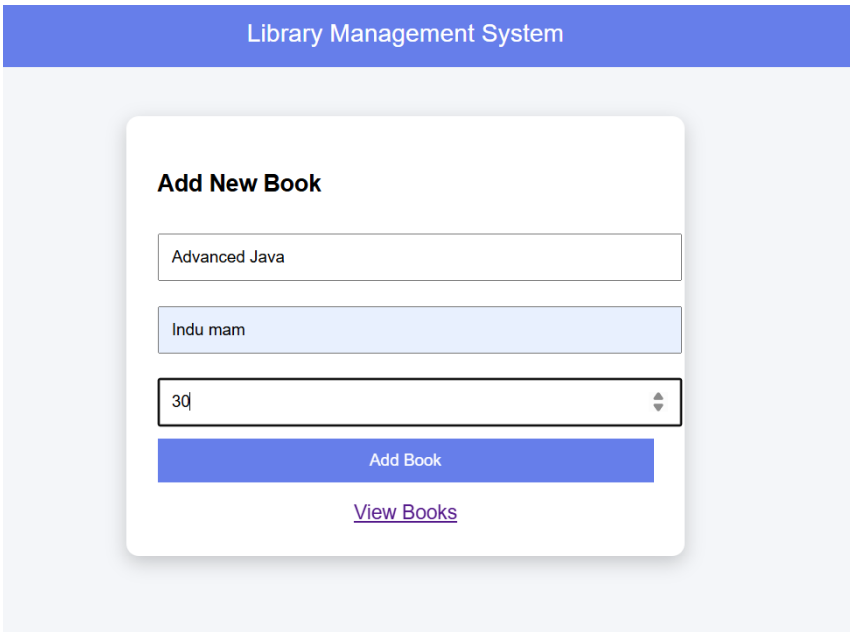
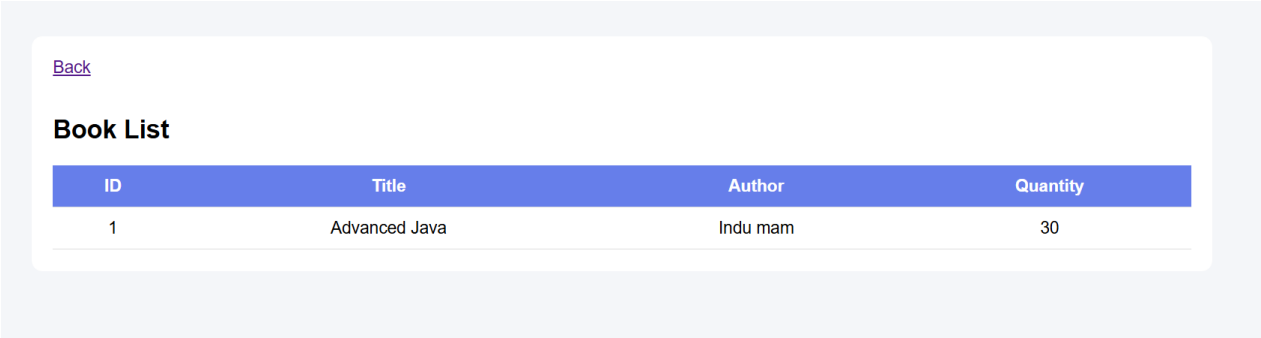
```
<%@ page import="java.util.*, model.Book" %>

<!DOCTYPE html>
<html>
<head>
    <title>View Books</title>
    <style>
        body {
            font-family: Arial;
            background: #f4f6f9;
        }
        .container {
            width: 80%;
            margin: auto;
            margin-top: 40px;
            background: white;
            padding: 20px;
            border-radius: 10px;
        }
        table {
            width: 100%;
            border-collapse: collapse;
        }
        th, td {
            padding: 10px;
            border-bottom: 1px solid #ddd;
            text-align: center;
        }
        th {
            background: #667eea;
            color: white;
        }
        tr:hover {
            background: #f1f1f1;
        }
```

```
        a {
            display: inline-block;
            margin-bottom: 15px;
        }
    </style>
</head>
<body>

<div class="container">

    <a href="addBook.jsp">⬅ Back</a>

    <h2>Book List</h2>

    <table>
        <tr>
            <th>ID</th>
            <th>Title</th>
            <th>Author</th>
            <th>Quantity</th>
        </tr>

        <%
            List<Book> list = (List<Book>) request.getAttribute("bookList");
            if(list != null){
                for(Book b : list){
        %>
        <tr>
            <td><%= b.getBookId() %></td>
            <td><%= b.getTitle() %></td>
            <td><%= b.getAuthor() %></td>
            <td><%= b.getQuantity() %></td>
        </tr>
        <%
                }
            }
        %>

    </table>

</div>

</body>
</html>
```

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Advanced Java (01CT1623)	**Aim:** Extend Task 1 to create a web-based Library Management System using Servlets	
Take Home Exam 2	**Date :-22-04-2026**	**Enrollment No:-** 92301733024

Servlet API Execution Flow

1. User sends request from browser
2. Request goes to Tomcat (Web Container)
3. Tomcat checks URL mapping
4. Corresponding Servlet is invoked
5. doGet()/doPost() method executes
6. Servlet interacts with DAO (JDBC)
7. Data is fetched from database
8. Data stored in request object
9. Request forwarded to JSP using RequestDispatcher
10. JSP generates response
11. Response sent back to browser

OUTPUT:

Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Advanced Java (01CT1623)	**Aim:** Extend Task 1 to create a web-based Library Management System using Servlets	
Take Home Exam 2	**Date :-22-04-2026**	**Enrollment No:-** 92301733024